RAMCO INSTITUTE OF TECHNOLOGY
Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology
Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: VI Semester, B.Tech. IT
Course Code & Title: CCS343 & Digital and Mobile Forensics
Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I **/** Digital Forensics Process
- **Course Outcome:** CO1
- **Topic Learning Outcome: TLO2**
- **Activity Chosen: Mind Map**

- **Justification:**
  The digital forensic process is a structured, four-to-five-step method designed to identify, preserve, analyze, and present digital evidence while ensuring integrity for legal proceedings. In Mind Map activity, the students will be given 5 minutes time to draw the conceptual diagram about the digital forensic process. Exploring this mind map provides a comprehensive overview of digital forensics. The objective of this activity is the process of generating, transmitting, and storing memory dumps within digital forensic

- **Time Allotted for the Activity: 05** Minutes

- **Details of the Implementation:**
  At the end of the lecture, students were instructed to draw a mind map for digital forensic process. Figure 1.a and Figure 1.b shows the mind maps drawn by S. Poongothai and D Pramila Devi. This task assesses students' understanding about forensic process in an easier way and considered as an excellent alternative to other checklist approaches. Most of the students presented unique maps based on their level of understanding.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	3	2	1	1	1	1	3	3	1	3	1

(1 – Low 2 – Moderate 3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	3	2	1	1	1	1	3	3	1	3	1
Justification for Correlation	The students could be able to use and apply basic mathematics knowledge and fundamental programming knowledge various digital forensics process.	The students could be able to identify basic engineering problem with various digital forensics process.	The students could be able to design the various design strategies for engineering problems	The students could be able to interpret the application of digital forensics	The students could be able to use various frameworks for implementing process of digital forensics	The students will complete the assessments in an ethical way.	The students will work as a team and as an individual. demonstrate communication, problem-solving skills.	The students will demonstrate communication, problem-solving skills.	The students will be able to identify the task required to collect the digital evidences	The student will identify the technological advances made in the past and update the concepts in digital forensics.	The students could be able to apply the computing skills for digital forensics.	The students could be able to provide IT solutions to the process of digital forensics.	The students could be able to develop software application for the digital forensics

- **Images / Screenshot of the practice:**

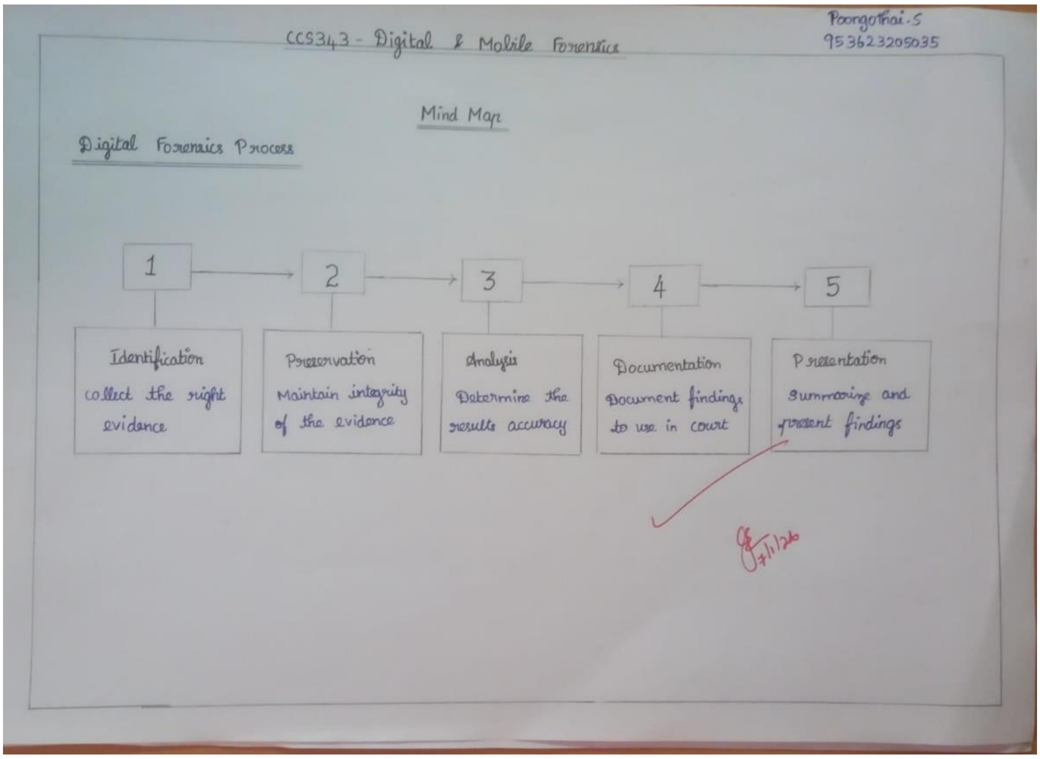

Figure 1. a Sample Mind Map of Poongothai S

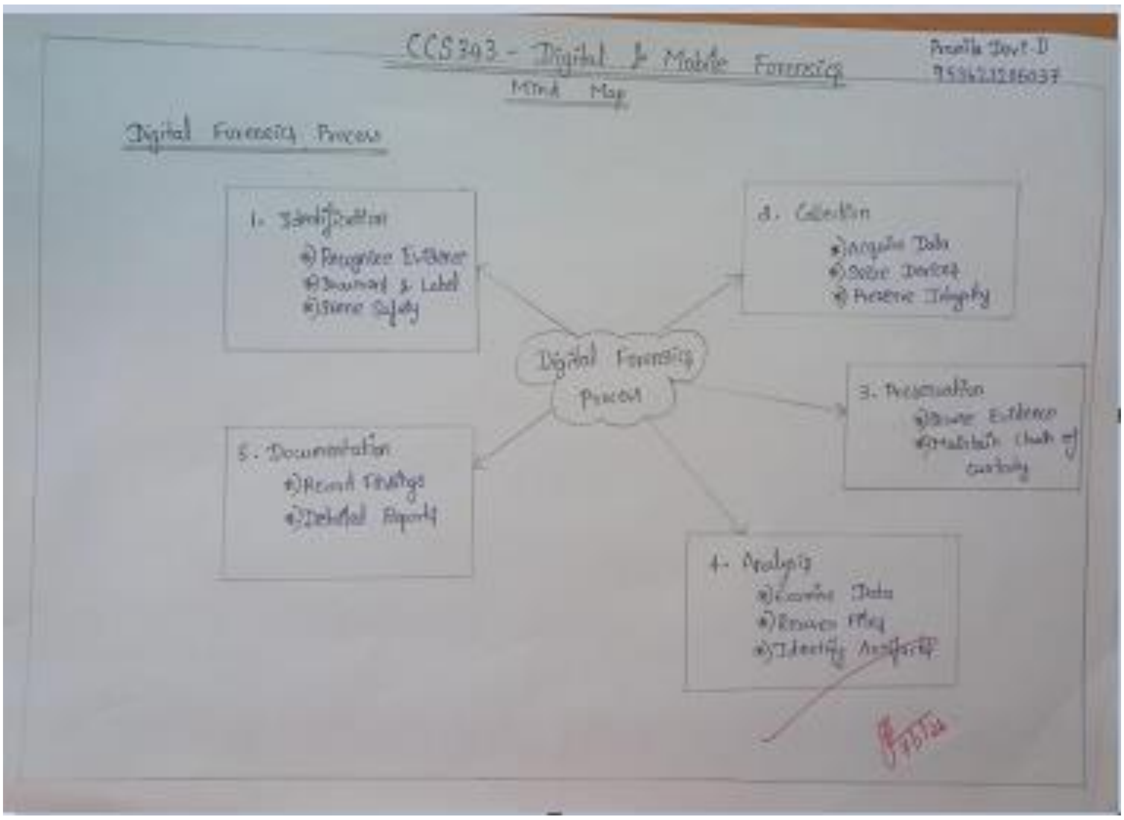

Figure 1. b Sample Mind Map of Pramila Devi D

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
    - Most of the students felt that this activity is highly interactive and engaging workshop method compared to static documents.
    - Stakeholders felt that this activity aids in defining how, when, and with whom to communicate, moving beyond one-size-fits-all strategies.
- ***Benefit of the practice:***
    - Students were actively participated in this activity and shown their excellence in creating varieties of mind map.
    - It acts as a cognitive tool that mimics the brain's associative, rather than linear, thinking process
    - By using keywords, colors, and images, mind maps create multiple, memorable, and visual cues that are easier for the students to recall.
- ***Challenges faced in implementation:***
    - For some students, the process of organizing and refining a complex map can be time-intensive, especially for beginners or when creating detailed maps.
    - Some students felt that the presentation of a mind map does not give them much clarity about the content. After final discussion, they got clear understanding and gained an idea for drawing the maps.

References:

- https://www.figma.com/resource-library/mind-map-examples/
- https://www.mural.co/blog/mind-mapping
- https://www.canva.com/graphs/mind-maps/
- https://venngage.com/blog/mind-map-examples/

Signature of Faculty Member **HOD**